

Ethical vs. Misleading Visualisation Reflection

Communicating Data Responsibly

This report examines how design choices in data visualisation influence interpretation, focusing on poverty and access to services. The aim is to demonstrate how ethical practices preserve accuracy, while misleading designs distort perception. By contrasting ethical and misleading versions of the same dataset, the analysis highlights why careful presentation is essential in business reporting and decision-making.

Ethical and Misleading Practices in Context

Visualisations serve as tools to reveal trends, yet poor design decisions can lead to misinterpretation. Ethical visualisation ensures proportional differences are retained, axes are scaled correctly, and context is transparent. Misleading visualisation, however, can arise from truncated axes, shortened timeframes, stacked proportions, or omitted legends. These distortions exaggerate or conceal patterns, shifting the narrative in ways that undermine trust and accuracy.

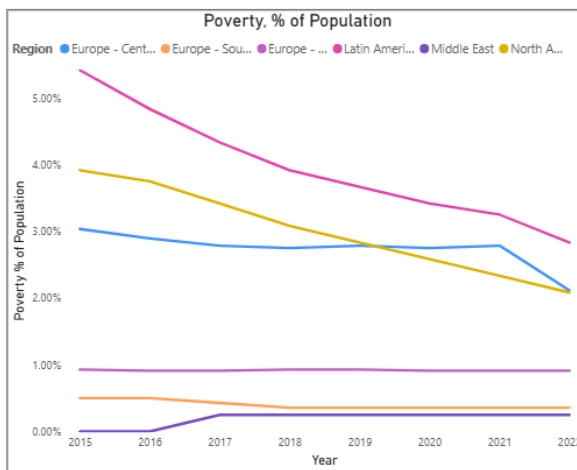


Fig 1.1 Ethical – Year wise trend

- ✓ Full period 2015–2022
- ✓ Axis starts at zero
- ✓ Legend on
- ✓ Proportional Differences

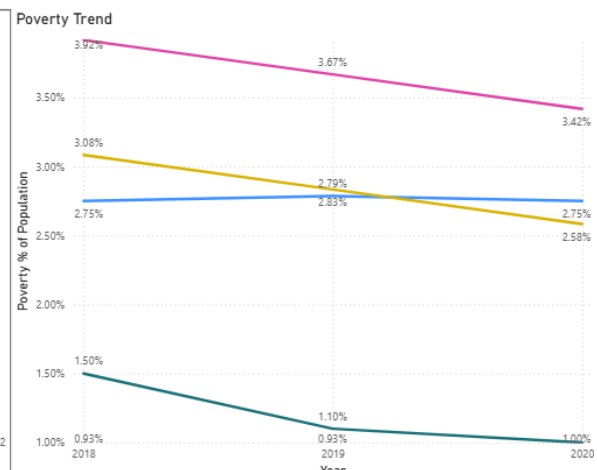


Fig 1.2 Misleading – Year wise trend

- ✗ Shortened period
- ✗ Axis starts at 1%
- ✗ Legend removed
- ✗ Exaggerates change

The ethical chart in Fig. 1.1 presents poverty trends from 2015 to 2022 with the Y-axis starting at zero and consistent regional labels. It shows a modest but steady decline, such as Latin America reducing from over 5% to around 3%, while Western Europe remains consistently below 1%. The presentation conveys both progress and disparity in proportional context. By contrast, the misleading version in Fig. 1.2 shortens the period to just a few years and shifts the Y-axis baseline to 1%. These adjustments exaggerate the slope of the lines, making gradual declines

appear dramatic. With the legend removed, regional context is obscured, leading viewers to believe changes are sharper and more volatile than they truly are.

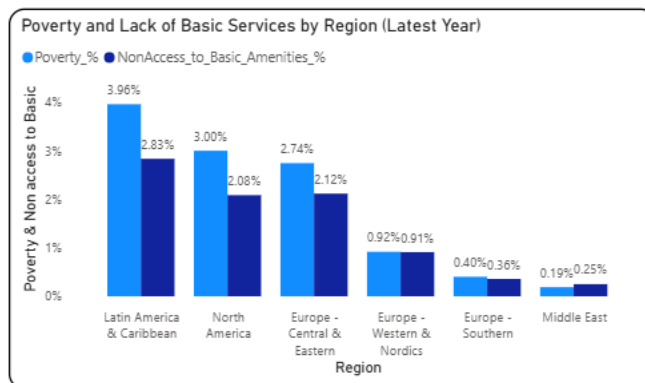


Fig 1.1 Ethical – Region wise

- ✓ Clustered to compare
- ✓ Axis Included, Sorted on values
- ✓ Shows true scale

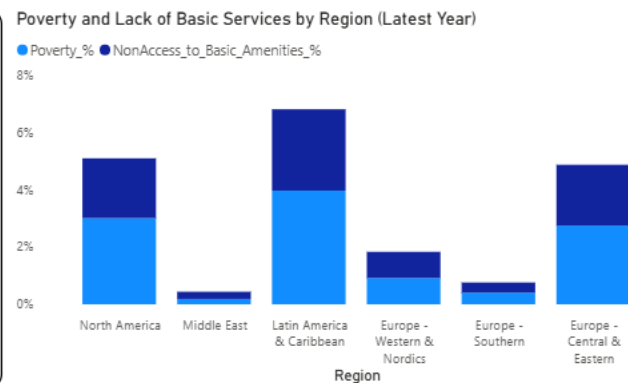


Fig 1.2 Misleading – Region wise

- ✗ 100% Stacked
- ✗ Odd sorting, No axis
- ✗ Hides Magnitude

Figure 2.1 provides an ethical comparison using clustered columns, where Poverty_% and NonAccess_to_Basic_Amenities_% are displayed with clear axes and precise data labels. This allows for transparent assessment, showing Latin America and North America with higher burdens compared to Europe or the Middle East. Magnitudes are preserved, and variation is visible. Figure 2.2 instead applies a 100% stacked column, equalizing all regions to the same bar height. This hides absolute differences, giving the impression that poverty and service access levels are similar across regions. Odd sorting and scale choices compound the distortion, flattening critical disparities.

Ensuring Accuracy Through Ethical Design

The comparison demonstrates that ethical practices build transparency and trust, while misleading designs risk confusion. Figures 1.1 and 2.1 highlight steady trends and real differences, while Figures 1.2 and 2.2 exaggerate or conceal change. Ethical visualisation ensures the data's message is preserved, enabling evidence-based decisions that respect both the data and its audience.